1 Indoor Air Quality Station Maintenance Manual

This manual provides a structured framework for maintaining indoor air quality monitoring stations across multiple sites. It covers routine procedures, calibration protocols, data validation, and quality assurance to ensure reliable and accurate measurements. The guidance is designed for facility managers, technicians, and environmental health officers responsible for station upkeep and data integrity.

Supporting records are presented in Figure 1.

Routine Maintenance Procedures



Figure 1: Daily inspection tasks and responsible roles

Routine maintenance is essential for consistent station performance. This section outlines daily, weekly, and monthly tasks, including sensor checks, filter replacements, and software updates. Adherence to these schedules minimizes downtime and ensures data continuity.

1.1.1 Daily Inspection Checklist

Daily inspections involve verifying power status, checking for physical obstructions, and confirming data transmission. Technicians should log any anomalies and escalate issues according to the response timeline. A structured checklist helps maintain consistency across all sites.

Supporting records are presented in Table 1.

Check	Acceptance	Action if fail	Record
Zero reading	Within tolerance	Recalibrate	Calibration log
Span reading	Within tolerance	Recalibrate	Calibration log
Data completeness	Near complete	Investigate	Data audit
Clock sync	Close match	Synchronize	Maintenance log

Table 1: Weekly verification criteria and actions

Sensor and Filter Checks

Sensors and filters require regular attention to prevent drift and contamination. This subsection details visual inspection of sensor heads, replacement intervals for particulate filters, and

methods to verify airflow. Proper handling and documentation are critical.

1.1.1.1.1 Particulate Sensor Cleaning

Particulate sensors accumulate dust, affecting accuracy. Cleaning with approved materials and following manufacturer guidelines is necessary. Technicians should record cleaning dates and observe post-cleaning readings to confirm sensor recovery.

Filter type	Interval	Condition	Responsible
Particulate	Quarterly	Visual	Technician
Gas	Semi-annual	Manufacturer	Technician
Pre-filter	Monthly	Visual	Technician

Table 2: Filter types and replacement intervals

1.1.1.1.2 Filter Replacement Schedule

Filter replacement follows a schedule based on operating hours and environmental conditions. The table below summarizes replacement intervals for different filter types, ensuring consistent maintenance across stations.

Supporting records are presented in Table 2.

Data Logging and Transmission

Reliable data logging and transmission are vital for continuous monitoring. This subsection covers checking logger status, verifying network connectivity, and ensuring time synchronization. Regular audits of data completeness help identify gaps early.

Supporting records are presented in Table 3.

Parameter	Check method	Frequency	Responsible
Logger status	Visual	Daily	Technician
Network connectivity	Ping test	Daily	Technician
Time sync	Compare	Weekly	Technician
Storage capacity	Review	Monthly	Supervisor

Table 3: Data logging parameters and checks

1.1.2 Weekly Performance Verification

Weekly verification includes comparing station readings with reference instruments and reviewing calibration status. Discrepancies beyond acceptable limits trigger recalibration or repair. Documentation of verification results supports quality assurance.

Documentation and Record Keeping

Accurate records of maintenance activities, calibration events, and data quality checks are essential. This section describes logbook formats, digital record systems, and retention policies to support audits and trend analysis.

2 Calibration and Quality Assurance

Calibration and quality assurance activities ensure measurement traceability and accuracy. This part details calibration frequency, reference standards, and corrective actions for out-of-tolerance results.

Calibration Protocols

Calibration protocols specify procedures for zero and span checks, use of certified reference materials, and adjustment of sensor responses. Technicians must follow documented steps and record all calibration data.

2.1.1 Calibration Frequency and Methods

Calibration frequency depends on sensor type and regulatory requirements. This subsection outlines recommended intervals and methods, including multipoint calibration for critical sensors and single-point checks for others.

Zero and Span Checks

Zero and span checks verify sensor baseline and response. Zero checks use clean air, while span checks employ known concentrations. Results outside acceptance criteria require immediate corrective action.

2.1.1.1.1 Reference Standards

Reference standards must be traceable and within validity. This subsection describes handling, storage, and verification of standard gases and permeation tubes to maintain their integrity.

Supporting records are presented in Table 4.

Standard	Storage	Shelf life	Verification
Zero gas	Ventilated	One year	Certificate
Span gas	Cool, dry	Half year	Certificate
Permeation tube	Constant temp	Short term	Weight check

Table 4: Reference standards handling requirements

Field	Description	Example	Required
Date	Calibration date	Recorded	Yes
Standard	Gas/tube ID	Cylinder tag	Yes
Result	Pass/fail	Pass	Yes
Technician	Name/ID	J. Doe	Yes

Table 5: Calibration record fields

2.1.1.1.2 Calibration Records

Calibration records include dates, standards used, results, and technician signatures. These records are retained for audit purposes and to track sensor performance over time.

Supporting records are presented in Table 5.

Corrective Actions for Drift

When drift is detected, technicians must investigate root causes, adjust or replace sensors, and document actions. Follow-up checks ensure the issue is resolved and data quality is restored.

2.1.2 Data Validation and Reporting

Data validation involves screening for outliers, flagging invalid periods, and applying correction factors. Validated data are compiled into reports for stakeholders, ensuring transparency and reliability.

Audit and Review Procedures

Regular audits of maintenance and calibration records ensure compliance with protocols. Review procedures include internal checks and periodic external assessments to identify areas for improvement.